QUESTIONS 2 / 10 completed

Binary Search

BINARY SEARCH

"Write a C program implementing binary search on a sorted array of 15 integers. The program must: (1) accept an unsorted array and sort it first, (2) perform binary search, (3) display search iterations with mid-point calculations, (4) show final result with comparison count,

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main()
{
  int n=15,a[15],i,j,temp,key;
  int low,high,mid,comp=0,found=0;
  for(i=0;i<n;i++)
  scanf("%d",&a[i]);
 for(i=0;i<n-1;i++)
   for(j=0;j<n-i-1;j++)
     if(a[j]>a[j+1])
     {
       temp-a[j];
       a[j]=a[j+1];
       a[j+1]=temp;
     }
  printf("soorted array:");
  for(i=0;i<n;i++)
    printf("%d",a[i]);
  printf("\nenter target: ");
  scanf("%d",&key);
  low=0;
```

OUTPUT

```
aorted array:51015202530354045505560708090
enter target: binary search iteration:
low=0, high=14, mid=7, arr[40]=268501009
low=8, high=14, mid=11, arr[60]=268501009
low=8, high=10, mid=9, arr[50]=268501009
low=8, high=8, mid=8, arr[45]=268501009
```

INPUT

50 30 10 90 40 70 20 80 60 5 45 35 25 55 15 45

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DS PROGRAMMING

H Harshavardhan
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QUESTIONS 2 / 10 completed

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PROGRAM EDITOR

C

Run Save

```
low=0;
high=n-1;
printf("binary search iteration:\n");
while(low<=high)
{
  mid=(low+high)/2;
  comp++;
  printf("low=%d, high=%d, mid=%d, arr[%d]=%d\n",low,high,mid,a[mid]);
  if(a[mid]==key)
  {
    found=1;

    break;

  }
  else if(a[mid]<key)
    low=mid+1;
  else
    high=mid-1;
}
if(found)
```

OUTPUT

```
aorted array:51015202530354045505560708090
enter target: binary search iteration:
low=0, high=14, mid=7, arr[40]=268501009
low=8, high=14, mid=11, arr[60]=268501009
low=8, high=10, mid=9, arr[50]=268501009
low=8, high=8, mid=8, arr[45]=268501009
```

INPUT

50 30 10 90 40 70 20 80 60 5 45 35 25 55 15 45

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PROGRAM EDITOR

C

Run Save

```
    printf("low=%d, high=%d, mid=%d, arr[%d]=%d\n",low,high,mid,a[mid]);
    if(a[mid]--key)
    {
      found=1;

      break;

    }
    else if(a[mid]<key)
      low=mid+1;
    else
      high=mid-1;
  }
  if(found)
    printf("found at index %d(1-based: %d)\n",mid,mid+1);
  else
    printf("element not found\n");
  printf("comparisons: %d\n",comp);
  printf("array remains sorted:\nyes");
  return 0;
}
```

OUTPUT

```
aorted array:51015202530354045505560708090
enter target: binary search iteration:
low=0, high=14, mid=7, arr[40]=268501009
low=8, high=14, mid=11, arr[60]=268501009
low=8, high=10, mid=9, arr[50]=268501009
low=8, high=8, mid=8, arr[45]=268501009
found at index 8(1-based: 9)
```

INPUT

50 30 10 90 40 70 20
80 60 5 45 35 25 55
15 45